Information Asymmetry

Higher-Level Psychological Descriptions as Compressions of Deeper Dynamical Representations

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Resurrection Tech · Morrison Framework · Version 2.1

"What survives when psychology is removed?"

Abstract

Psychological concepts such as anxiety, confidence, motivation, attention, and emotion provide compact, communicable descriptions of behaviour. This paper develops and operationalises a representational hypothesis: *higher-level psychological descriptions are compressions of deeper dynamical descriptions*. The claim is neither eliminativist nor a denial of the practical value of psychological language. It predicts a directional information asymmetry. A sufficiently informative dynamical representation should support prediction of psychological descriptions, whereas those descriptions alone should generally fail to reconstruct the trajectory geometry, transition structure, constraints, feedback, reachability, and intervention-relevant information from which they were derived. We distinguish the hypothesis from adjacent work in dynamical cognition, information bottleneck theory, predictive-state representations, psychometrics, and control theory; formalise forward prediction, reverse reconstruction, predictive remainder, and intervention specificity; and propose a staged, preregistered experimental programme with capacity-matched baselines, perturbations, negative controls, out-of-distribution validation, and explicit falsification criteria. We also describe Runtime Governance, an operational engineering implementation inspired by the same structural vocabulary. Its bounded engineering results show that structural representations can be useful for pre-execution governance in autonomous systems. They do not test, and are not presented as proof of, the scientific hypothesis. The contribution is therefore a testable research programme: a precise claim about representational depth, a method for measuring the predicted asymmetry, and a disciplined separation between scientific inference and engineering demonstration.

Keywords. Dynamical systems · psychological description · information asymmetry · lossy compression · state space · reachability · predictive sufficiency · runtime governance · early detection.

1 Motivation

Psychology has developed powerful descriptive concepts for organising behaviour. A word such as *anxiety* can condense reports, bodily changes, attentional patterns, avoidance, temporal expectations, and social meaning into a form that can be communicated and acted upon. The usefulness of that description does not, by itself, determine its representational depth. A dashboard can be useful precisely because it hides most of the engine.

> *Are psychological descriptions the most fundamental representation of behaviour, or are they compressions of deeper dynamical structure?*

The proposal is the second possibility. On this view, a psychological description remains real and useful at its own level, but it may be a many-to-one projection of a richer representation containing state, trajectory, constraint, feedback, transition, reachability, and control information. Calling the description a compression is an empirical claim about information retention, not a rhetorical judgment about the legitimacy of psychology.

Why the representational direction matters

If a dynamical description can generate a psychological label, predict relevant outcomes beyond that label, and identify different intervention requirements within the same label class, then it contains distinctions the label has discarded. Conversely, if the label reconstructs the relevant dynamics and is sufficient for prediction and intervention, the claim of deeper retained structure is weakened.

- **Explanation.** A representation that specifies how behaviour evolves under perturbation may explain more than one that only names the observed regime.
- **Prediction.** Dynamical variables can, in principle, reveal approach toward a transition before a familiar descriptive threshold is crossed.
- **Intervention.** Two systems carrying the same label may occupy different states, have different reachable futures, and require different controls.
- **Measurement.** Compression can be operationalised through reconstruction loss, conditional information, predictive performance, and model description length.
- **Generalisation.** State, transition, constraint, feedback, and reachability are available across engineered, biological, and social systems even when domain-specific labels differ.

None of these consequences is assumed to hold universally. The purpose of the programme is to make the proposed asymmetry measurable and therefore vulnerable to refutation.

2 Scope, Terms, and Claim Boundaries

The word *deeper* can be misleading unless constrained. Here it means that a representation retains finer temporal, relational, counterfactual, or intervention-relevant distinctions. It does not mean metaphysically privileged, morally superior, or necessarily closer to a particular biological substrate.

Term	Operational meaning	Not assumed
Psychological description	A label, rating, construct score, or structured report used to summarise an episode or behavioural regime.	That every construct is vague, invalid, or reducible to one mechanism.
Dynamical representation	A temporally indexed model of states, observations, actions, transitions, feedback, constraints, and reachable futures.	That the true state is fully observed or that one model class is uniquely correct.
Compression	A mapping that preserves some distinctions while discarding others relative to a target or distortion function.	That compression is necessarily harmful or that shorter language is less useful.
Deeper	Richer with respect to preregistered structural or predictive targets.	An unrestricted claim of ontological priority.
Generate	Predict a psychological description from independently constructed dynamical information on held-out data.	Mere post-hoc relabelling or training and testing on the same episodes.
Reconstruct	Recover preregistered dynamical properties from psychological descriptions alone.	Producing any plausible trajectory compatible with a label.

Three objects must remain separate

– **Scientific hypothesis.** A claim about the relation between psychological and dynamical representations.
– **Proposed experiment.** A family of tests designed to estimate that relation and identify conditions under which it fails.
– **Engineering implementation.** Runtime Governance, which applies related structural concepts to autonomous-system execution. It demonstrates engineering utility in its defined domain, not the truth of the scientific hypothesis.

This separation prevents two invalid inferences: successful engineering does not establish a theory of human behaviour, and a future failure of the psychological-compression hypothesis would not by itself invalidate an independently tested governance mechanism.

3 Background and Literature Context

The hypothesis sits at the intersection of several traditions. None states the present claim in the form tested here; each supplies part of the conceptual or methodological foundation.

Information theory, lossy coding, and relevance

Shannon's communication theory made uncertainty and information loss measurable (Shannon, 1948). Rate-distortion theory later formalised the trade-off between representational rate and tolerated distortion (Berger, 1971). The information bottleneck method reframed compression around relevance: a compact representation should discard information about an input while preserving information useful for predicting a target (Tishby, Pereira, and Bialek, 1999). Minimum-description-length approaches similarly connect explanation to compact coding under an explicit model class (Rissanen, 1978).

These frameworks make an important point for the present paper: compression is always relative to what must be preserved. A psychological label may be highly efficient for communication or coarse decision making while being insufficient for reconstructing transition topology or selecting a precise intervention. The hypothesis therefore does not predict that psychological descriptions preserve *no* information. It predicts that relevant dynamical information remains after conditioning on them.

Dynamical approaches to cognition and behaviour

Cybernetics and general systems theory foregrounded feedback, regulation, and organisation (Wiener, 1948; Ashby, 1956; von Bertalanffy, 1968). Later dynamical approaches treated cognition and behaviour as time-evolving systems rather than sequences of static symbolic states (van Gelder, 1998; Beer, 2000). Coordination dynamics showed how collective variables, attractors, and phase transitions can describe behavioural organisation (Haken, Kelso, and Bunz, 1985; Kelso, 1995), while developmental systems approaches emphasised behaviour emerging across interacting timescales (Thelen and Smith, 1994).

The present hypothesis is compatible with this lineage but more directional. It does not merely claim that behaviour can be modelled dynamically. It predicts an asymmetry between what a dynamical representation can produce and what a psychological description can recover.

State, prediction, and control

State-space models represent a system through variables sufficient to determine, probabilistically or deterministically, its subsequent evolution under inputs. Kalman's filtering framework formalised estimation of hidden state from noisy observations (Kalman, 1960). Predictive-state representations showed that state can also be expressed through action-conditional predictions of future observations (Littman, Sutton, and Singh, 2002). Reachability and viability theory ask which states can occur under admissible controls and constraints (Aubin, 1991; Mitchell, Bayen, and Tomlin, 2005).

These concepts sharpen the paper's use of *structure*. A label may describe the present surface while omitting where the system can go, which transitions are feasible, how it responds to perturbation, and which actions move it away from or toward a boundary.

Psychological constructs and levels of explanation

Construct validation distinguishes a theoretical construct from any single observation or instrument (Cronbach and Meehl, 1955), and latent-variable work makes explicit that statistical constructs require substantive interpretation rather than automatic reification (Borsboom, Mellenbergh, and van Heerden, 2003). Marr's levels of analysis likewise caution against collapsing computational purpose, representation, and physical implementation into one description (Marr, 1982).

Accordingly, this paper does not argue that psychological constructs are merely words. A validated construct can support reliable prediction and coordination. The narrower question is whether it is a sufficient statistic for the dynamical properties relevant to future behaviour and intervention.

The research gap

Existing work provides theories of compression, dynamical models of behaviour, state estimation, and construct validity. What remains to be tested directly is the proposed mapping asymmetry:

> A richer dynamical representation should generate or predict psychological descriptions, while those descriptions should not fully reconstruct the original dynamics or exhaust their predictive and intervention-relevant information.

4 Core Hypothesis

> **Hypothesis.** Higher-level psychological descriptions are compressions of deeper dynamical descriptions.
>
> If correct: dynamics should *generate* psychology, while psychology should not completely *reconstruct* dynamics. This produces an observable information asymmetry.

The wording above is the hypothesis. The formal statements that follow are operational consequences, not replacements for it. Psychological language remains useful; the claim is only that it may not preserve every structural distinction relevant to prediction and intervention.

Consider an episode indexed by $t = 0, \ldots, T$. Let $x_t \in \mathcal{X}$ be a latent or operational state, $u_t \in \mathcal{U}$ an input or action, and $y_t \in \mathcal{Y}$ an observation:

$$x_{t+1} = F_\theta(x_t, u_t) + \xi_t, \qquad y_t = G_\theta(x_t) + \eta_t,$$

where F_θ and G_θ describe transition and observation relations, and ξ_t, η_t represent process and measurement uncertainty. A dynamical representation is

$$\mathcal{D} = (\mathcal{X}, \mathcal{U}, \mathcal{Y}, x_0, F_\theta, G_\theta, C, \mathcal{R}, \Omega, \tau),$$

where C denotes constraints, $\mathcal{R}$ reachable states, Ω a task-defined adverse or forbidden region when one exists, and $\tau = (x_{0:T}, u_{0:T-1}, y_{0:T})$ the episode trajectory. Let P be a psychological description, potentially multivariate and noisy.

The hypothesised relation is represented by a forward map and a reverse estimator:

$$f_\phi : \mathcal{D} \to P, \qquad g_\psi : P \to \widehat{\mathcal{D}}.$$

Three empirical components follow.

- **Forward expressivity.** There exists a model f_ϕ that predicts P from independently constructed features of $\mathcal{D}$ on held-out episodes and, ideally, across controlled distribution shifts.
- **Reverse underdetermination.** Even an optimised g_ψ leaves preregistered structural features of $\mathcal{D}$ unrecovered because multiple distinguishable dynamics map to the same description.
- **Predictive remainder.** For relevant future, perturbational, or intervention targets Z, dynamical information remains after the psychological description is known:

$$I(\mathcal{D}; Z \mid P) > 0.$$

Forward prediction alone is insufficient. A high-capacity model could predict a label from almost any correlated data. The hypothesis is supported only by the joint pattern: reliable forward mapping, demonstrable reverse loss, and retained dynamical value under controlled comparisons.

5 Information Asymmetry

If psychology is a compression of dynamics, the two directions of mapping are not equivalent. The forward direction should preserve information; the reverse should not.

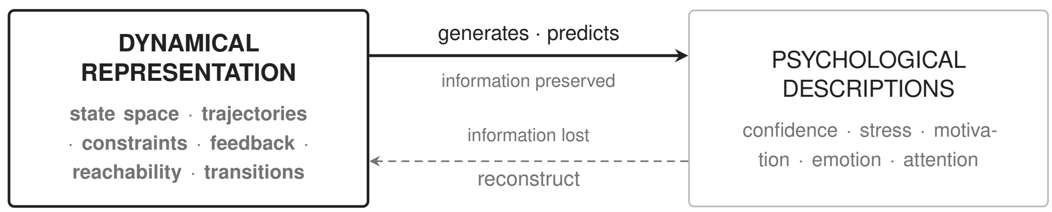


Figure 1. The hypothesised directional asymmetry. The solid mapping predicts a compact psychological description from dynamics. The dashed mapping attempts to reconstruct dynamics from that description alone and is expected to remain underdetermined.

Many-to-one structure

For a description p, define its dynamical equivalence class

$$[\mathcal{D}]_p = \{\mathcal{D}_i : f_\phi(\mathcal{D}_i) = p\}.$$

If this set contains dynamically distinct members, then knowing p does not identify the originating dynamics. The distinction must be task-relevant: members should differ in a preregistered property such as transition topology, basin membership, distance to a boundary, response to perturbation, or optimal intervention.

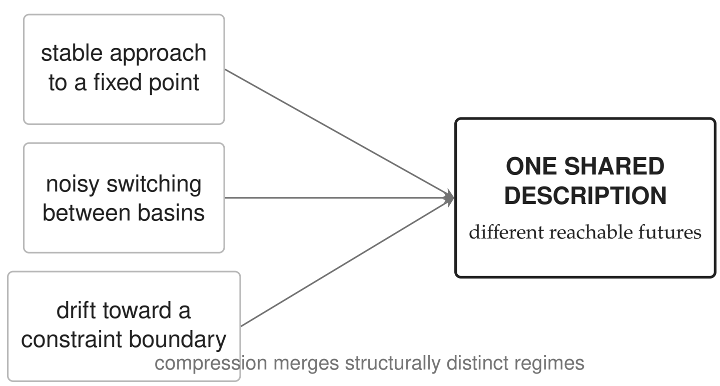


Figure 2. Many-to-one compression. A shared description can remain communicatively useful while merging dynamical regimes with different futures and intervention requirements. The examples are schematic, not claims about a specific construct.

6 Information-Theoretic Interpretation

In an idealised discrete setting, if P is generated from $\mathcal{D}$ and retains less than the full representation, then

$$H(\mathcal{D} \mid P) > 0, \qquad I(\mathcal{D}; P) < H(\mathcal{D}).$$

The first term measures residual uncertainty about the dynamics after the description is known. The second states that the description does not carry all information present in the dynamical representation.

For a target Z, define the *predictive remainder*

$$\Gamma_Z = I(\mathcal{D}; Z \mid P).$$

A positive Γ_Z means that the dynamical representation contains target-relevant information not exhausted by the psychological description. Targets may include future observations, transition events, responses to perturbation, or the outcome of a candidate intervention.

The reverse reconstruction objective is

$$L^*_{\text{rev}} = \inf_{g \in \mathcal{G}} \mathbb{E}\big[d_{\mathcal{D}}(\mathcal{D}, g(P))\big],$$

where $d_{\mathcal{D}}$ is specified before analysis. It should not be a single generic Euclidean error if the relevant structure is topological or intervention-based. A composite preregistered distortion can include trajectory, transition, reachability, constraint, and intervention terms.

Compression is target-relative

Rate-distortion and information-bottleneck theory prevent an overly simple conclusion. A representation can discard enormous detail and still be optimal for a target. Thus, the relevant question is not merely whether P loses information. Almost every abstraction does. The stronger question is:

> Does P discard information needed to explain, predict, distinguish, or intervene on the outcomes that the psychological description is used to address?

If the answer is no for the preregistered targets, the hypothesis receives little support in that domain. If the answer is yes only because $\mathcal{D}$ has more parameters, poorer regularisation, or access to future information, the result is confounded.

Continuous variables and estimation

Human and engineered trajectories are often continuous, partially observed, and nonstationary. Differential entropy can be coordinate-sensitive and mutual-information estimators can be unstable in high dimensions. The primary analysis should therefore combine several operational measures: held-out log loss, proper scoring rules, minimum description length, calibrated prediction, topology-aware reconstruction, and perturbational tests. Information-theoretic quantities are guiding constructs; no single estimator should carry the conclusion.

7 Dynamical Systems Interpretation

A dynamical representation describes how a system changes, not merely what category it currently resembles. Relevant structures include fixed points and attractors, transient trajectories, basin boundaries, metastable regimes, bifurcations, feedback gain, controllability, observability, and reachable sets.

Three distinctions are central.

- **State versus appearance.** Similar observations can arise from different latent states, and similar states can produce different observations under different contexts.
- **Trajectory versus snapshot.** Direction, velocity, recurrence, and proximity to a transition cannot generally be inferred from a single label.
- **Reachability versus likelihood.** A low-probability future may still matter if it is reachable and catastrophic; a likely future may be avoidable under a small control.

Shared surface	Possible dynamical distinction	Why it matters
Same stress rating	One episode is mean-reverting; another is drifting toward a boundary.	Equal present scores can imply unequal near-term risk.
Same confidence label	One regime is robust to perturbation; another depends on fragile positive feedback.	A perturbation may reveal different stability and recovery.
Same attention score	One trajectory alternates between metastable states; another remains diffuse and noisy.	Similar averages can conceal different temporal organisation.
Same motivation report	One system is control-limited; another is constraint-limited.	The effective intervention may differ even when the description matches.

The entries are intuitions, not empirical findings. Their role is to specify what a future experiment must demonstrate: not merely different time series under the same word, but differences that survive controls and alter prediction or intervention.

Observability and reconstruction

Reconstructing state from observations is itself a difficult inverse problem. Delay-embedding results show that under strong conditions aspects of an attractor can be reconstructed from time-delayed measurements (Takens, 1981), but those conditions do not imply that a coarse psychological label identifies a unique behavioural state. The experiment must therefore distinguish loss caused by the proposed compression from loss caused by inadequate sensors or a poor state estimator.

Early detection

If a harmful transition is preceded by measurable structural change, quantities such as distance to a boundary, recovery time after perturbation, increasing autocorrelation, variance, or loss of resilience may provide earlier warning (Scheffer et al., 2009). Such indicators are neither universal nor automatically causal. The paper predicts only that, in systems where pre-transition structure exists, a dynamical representation has the opportunity to detect it before a thresholded description.

8 Proposed Experimental Test

The hypothesis should be tested in stages. Beginning with a broad clinical or societal claim would mix representational failure with measurement noise, ethical constraints, hidden confounding, and nonstationarity. A stronger sequence moves from known dynamics to controlled human behaviour and only then to external domains.

Stage A: pipeline validation with known dynamics

Generate families of controlled dynamical systems with known states, transitions, constraints, and perturbation responses. Construct deliberately coarse descriptions using transparent label functions, then test whether the analysis recovers the known forward mapping and known reverse loss. This validates estimators and falsification logic; because the labels are assigned rather than psychologically validated, it does not test the central hypothesis.

Stage B: controlled behavioural experiment

Use a continuous task in which participants or agents act under observable constraints and controlled perturbations. Candidate paradigms include sensorimotor coordination, adaptive pursuit, sequential resource allocation, or navigation under changing control costs. Each episode should provide dense behavioural observations, controlled inputs, independently collected psychological descriptions, held-out targets, and repeated episodes sufficient to separate within-system from between-system variation.

The dynamical representation must be learned without access to the psychological label. Candidate models should include transparent state-space baselines as well as more flexible nonlinear alternatives. Model selection must occur inside the training folds.

Stage C: external and out-of-distribution validation

Replicate on a second task, population, measurement instrument, and perturbation regime. The strongest evidence would be a structural representation whose predictive remainder and intervention distinctions persist when familiar label correlations shift.

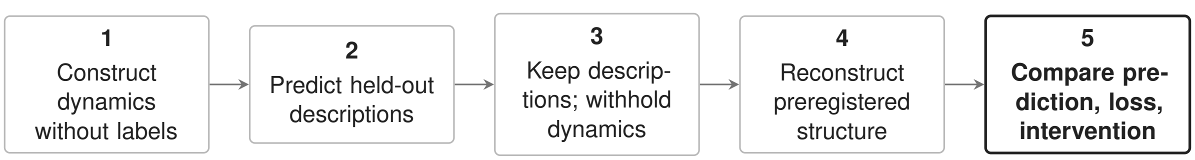


Figure 3. The core protocol. The label firewall between stages 2 and 3 prevents reverse reconstruction from using the original dynamics. Targets, distances, model classes, and thresholds are preregistered.

Forward test

Estimate $q_\phi(P \mid \mathcal{D})$ using nested cross-validation grouped by participant or system. Report proper scoring rules, calibration, uncertainty intervals, and out-of-distribution performance. Compare against label base rates and static summaries with matched parameter budgets. Success requires more than above-chance classification: it should be stable across held-out systems and not depend on an identity shortcut.

Reverse test

Train g_ψ using only P and permitted context variables. Reconstruct separately:

- trajectory class and temporal order;
- transition graph or local flow;
- basin or regime membership;
- constraint and boundary distances;
- perturbation response;
- reachable-set properties;
- future-event distribution;
- intervention ranking.

The reverse estimator should be strong, not deliberately weak. Bayesian and retrieval-based decoders can expose multimodality rather than forcing one reconstruction. The test concerns unavoidable underdetermination after reasonable modelling effort.

Predictive remainder and intervention test

For each target Z, compare

$$M_P : Z \sim P, \qquad M_D : Z \sim \mathcal{D}, \qquad M_{P+D} : Z \sim (P, \mathcal{D}).$$

The incremental value of dynamics is estimated by the held-out improvement of M_{P+D} over M_P, with capacity, regularisation, and data access matched as closely as possible. For interventions, test whether episodes sharing P require different actions to achieve the same preregistered outcome. Whenever ethically and practically possible, controlled perturbations should verify predicted differences rather than relying only on observational association.

Baselines and negative controls

Comparison	Purpose	Failure it detects
Static feature baseline	Match the number and type of non-temporal measurements.	Apparent gains caused by more variables rather than dynamics.
Capacity-matched label embedding	Give the label model comparable parameters and regularisation.	Model-capacity advantage.
Time-shuffled trajectory	Preserve marginals while destroying temporal order.	Gains unrelated to temporal structure.
Future-leakage audit	Verify every feature exists at the prediction time.	Artificial early warning.
Instrument replication	Repeat with a second psychological measure.	A result specific to one noisy scale or rater.
System-held-out split	Hold out entire individuals or agents.	Memorisation of identity rather than transferable structure.
Perturbational check	Apply a controlled input and observe recovery.	Purely correlational distinctions with no dynamical consequence.

Preregistration and statistical design

The confirmatory protocol should preregister the primary target, dynamical representation class, psychological instrument, distortion functions, exclusion rules, cross-validation grouping, stopping rule, and minimum effect considered theoretically meaningful. Sample size should be determined by simulation using the hierarchical dependence structure, not by treating time points as independent observations. Uncertainty should be estimated at the system or participant level.

9 Falsifiable Predictions and Outcome Patterns

The hypothesis is strengthened by a coordinated pattern, not a single favourable metric.

ID	Prediction	Supportive observation	Observation that weakens the claim
P1	Forward adequacy	Dynamics predict held-out descriptions with calibration and transfer.	Performance collapses outside training identities or does not exceed static controls.
P2	Reverse loss	Labels fail to recover preregistered structural properties despite strong decoders.	Labels reconstruct relevant dynamics within the equivalence margin.
P3	Predictive remainder	Dynamics improve future or perturbation prediction conditional on labels.	M_{P+D} does not improve meaningfully over M_P.
P4	Intervention specificity	Same-label episodes require different controls predicted by dynamics.	Label-conditioned interventions perform as well as state-conditioned interventions.
P5	Earlier detection	Dynamical indicators provide verified lead time before a descriptive threshold.	Lead time disappears after leakage, capacity, and base-rate controls.
P6	Partial invariance	Key asymmetries replicate across instruments or tasks.	The effect is tied to one label definition, site, or task.

Supportive, null, and mixed outcomes

A supportive result would show accurate forward prediction, materially imperfect reverse recovery, positive incremental prediction for preregistered targets, and at least one perturbational or intervention consequence. The claim would remain domain-bound until replicated.

A clean null is informative. If a well-validated psychological description is effectively sufficient for future prediction and intervention within the tested domain, then the hypothesised deeper advantage is not observed there. The programme should report equivalence bounds rather than interpreting non-significance as automatic support for either side.

The most plausible early result may be selective asymmetry: some constructs may behave as strong task-relevant compressions, while others merge dynamically distinct regimes. The hypothesis can then be refined by specifying the conditions, targets, and timescales on which it holds. Refinement is not license to evade falsification; boundary conditions must be stated prospectively as evidence accumulates.

10 Alternative Explanations and Limitations

Representational richness and temporal resolution

The dynamical model will usually receive more data than a label. Better prediction may therefore reflect input bandwidth rather than a uniquely dynamical advantage. Capacity-matched static summaries, compression-matched dynamic codes, downsampled histories, and time-shuffled controls are essential.

Label noise and construct mismatch

Reverse loss can arise because a scale is unreliable, a rater is inconsistent, or the label addresses a different target. This would show that a particular measurement is lossy, not that psychological descriptions in general are compressions of dynamics. Multiple instruments and latent measurement models should separate construct variance from measurement error.

State-estimation error

The proposed deeper representation is itself inferred. If sensors miss important variables or the model class is wrong, $\mathcal{D}$ may be a poor approximation. Forward success does not make it the true dynamics. Results must be compared across plausible state representations and accompanied by posterior or ensemble uncertainty.

Multiple realisability

Several microdynamics can support the same higher-level function. This may explain reverse underdetermination without implying that the lower-level description is always better for the target. In some settings, the macrodescription may be more stable, transportable, and causally informative. The paper therefore evaluates task-relative sufficiency rather than equating detail with explanation.

Prediction, context, and ethics

Conditional information and forecast accuracy do not identify causal mechanisms. Intervention claims require randomised or otherwise identified perturbations. Psychological and dynamical mappings may also vary across individuals, cultures, tasks, developmental stages, and measurement contexts. Hierarchical models and explicit distribution-shift tests should distinguish universality from local validity.

Early-warning models applied to health or behaviour can create false reassurance, stigma, or coercive intervention. The research programme does not authorise clinical use. High-stakes deployment would require independent validation, governance, informed consent, error-cost analysis, and domain-specific oversight.

Interpretive limit. Evidence that dynamics outperform a psychological description for one target does not show that psychology is useless, that subjective experience is unreal, or that all psychological constructs reduce to one dynamical vocabulary.

11 Engineering Inspiration

The hypothesis motivated an engineering implementation built on structural dynamics rather than psychological concepts. Instead of asking whether an autonomous system is confident, deceptive, or anxious, the implementation evaluates proposed actions, transitions, data flows, constraints, feasible continuations, and forbidden reachable outcomes before execution.

The engineering domain is attractive methodologically because proposed tool calls, arguments, permissions, and outcomes can be represented explicitly. Yet the analogy has limits. An autonomous workflow is not a model of a human mind, and a deterministic governance verdict is not evidence about psychological compression.

12 Runtime Governance

Runtime Governance is a pre-execution control layer for autonomous systems. It receives a proposed plan or tool-call trajectory, canonicalises the actions into an evaluable representation, applies domain-specific constraints and forbidden sets, estimates reachable consequences within its defined horizon, and returns a verdict before the plan is allowed to interact with production systems.

Local and global safety invariants

For one represented environment, the Morrison Safety Invariant is

$$\mathcal{R}(t) \cap \Omega = \varnothing$$

The reachable state set at time t has empty intersection with the forbidden region.

The global form requires the same separation across every environment in the bounded admissible set:

$$\text{Safe} \iff \forall E \in \mathcal{E}, \quad \mathcal{R}_E(t) \cap \Omega = \varnothing$$

Safety is a global property of reachable trajectories, not a judgment about an isolated output.

In both expressions, t is an ordinary math-italic time variable. The first equation is local to one represented environment. The second lifts the invariant across the admissible environment set $\mathcal{E}$. In practice, $\mathcal{R}_E(t)$ is approximated through bounded sampling or forward projection over the declared horizon.

Symbol	Meaning
t	Time or evaluation-step variable, typeset in mathematical italic.
$\mathcal{R}(t)$	Reachable state set at time t in one represented environment.
E, $\mathcal{E}$	A specific environment and the bounded set of admissible environments.
$\mathcal{R}_E(t)$	Reachable state set at time t under environment E.
Ω	Forbidden region of state space, defined by the applicable constraints.
$\cap$, $\varnothing$	Set intersection and the empty set.

Conditional safety guarantee

Let $K_E = \mathcal{X}_E \setminus \Omega$ denote the feasible region. If $x_0 \in K_E$ and, at every transition step, the governance policy permits only actions whose successors remain in K_E for every $E \in \mathcal{E}$, then for every evaluated time t,

$$\mathcal{R}_E(t) \cap \Omega = \varnothing.$$

Proof sketch. The initial state lies outside Ω. By assumption, every permitted transition preserves membership in K_E. Induction over the transition sequence therefore keeps every constructed trajectory in K_E, so no reachable state enters Ω. The guarantee is conditional on the represented transition model, admissible environment set, policy, and evaluation horizon.

Committed-to-failure condition

The complementary condition identifies a state from which every admissible continuation eventually enters the forbidden region:

$$s \in \mathcal{C}_\Omega \iff \forall \tau \in \text{Traj}(s),\ \exists t \geq 0 :\ \tau(t) \in \Omega.$$

Such a state is committed to failure relative to the declared model and admissible trajectory set: failure is unavoidable regardless of the remaining policy choice. This definition is distinct from a state that is merely risky or likely to fail.

The invariants are engineering contracts. Their force depends on the completeness of the trajectory representation, environment model, horizon, semantic lifting, and forbidden-set specification. Unknown or malformed plans require explicit fail-closed or escalation behaviour rather than an inference of safety.

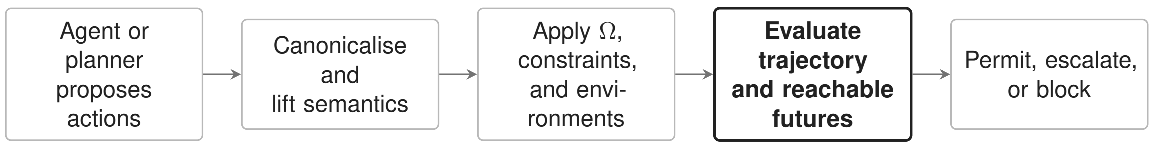


Figure 4. Runtime Governance at the execution boundary. The system governs a proposed action trajectory before tools run. Its safety claims remain scoped to the represented environment, policy, horizon, and tested failure modes.

Structural correspondence

Research concept	Engineering counterpart	Boundary
State and trajectory	Ordered tool calls, arguments, data-flow state, and execution context.	Discrete operational representation, not a complete world model.
Constraints	Permissions, schemas, policies, feasibility, and domain rules.	Only declared or inferable constraints can be enforced.
Reachability	Consequences reachable through proposed action chains.	Computed within an evaluation horizon and environment model.
Forbidden region Ω	Unauthorised transfer, data exfiltration, privilege escalation, or domain-specific prohibited outcome.	Domain definitions require independent specification and validation.
Feedback and governance	Verdicts, audit evidence, shadow mode, enforcement, and policy updates.	Organisational controls remain necessary around the runtime layer.

Operational demonstration

The system is publicly inspectable through a live demonstration and evidence page:

- **Live console —** resurrection-tech.com/live-demo
- **Evidence and methodology —** resurrection-tech.com/evidence
- **Assessment pathway —** resurrection-tech.com/assessment

These links connect the structural concepts to an operational engineering system. They do not provide observations of human psychological descriptions and therefore cannot test the central hypothesis.

13 Engineering Evidence and Evidential Boundary

The Runtime Governance implementation reports deterministic evaluation of proposed trajectories against defined forbidden sets, pre-execution interception, audit records, multi-domain policy definitions, and production onboarding. Publicly reported benchmark figures are explicitly scoped to internal, defined test suites rather than universal safety guarantees (Resurrection Tech, 2026b).

What the engineering evidence can establish	What it cannot establish
The implementation can encode and evaluate structural action representations in its supported domains.	That psychological descriptions are compressions of human or biological dynamics.
Specified unsafe trajectories can be blocked before execution in tested environments.	That every harmful trajectory, unknown environment, or adversarial encoding is covered.
Deterministic replay can produce repeatable verdicts for the same represented inputs and policy.	That the representation is complete, causally correct, or universally safe.
Structural vocabulary is sufficiently expressive for a functioning governance product.	That structural vocabulary is the most fundamental description in every scientific domain.

This boundary is not a weakness to be hidden. It is the condition that keeps the research programme falsifiable. Engineering motivates measurement and supplies implementation discipline; scientific evidence must come from the proposed experiments.

14 From Theory to Engineering to Programme

Theory, experiment, and engineering form a connected but non-circular arc. The hypothesis defines a representational claim; the experiment tests it; the engineering implementation demonstrates that related structural representations are practically expressive; and the programme explores whether comparable measurements remain useful in progressively less controlled systems.

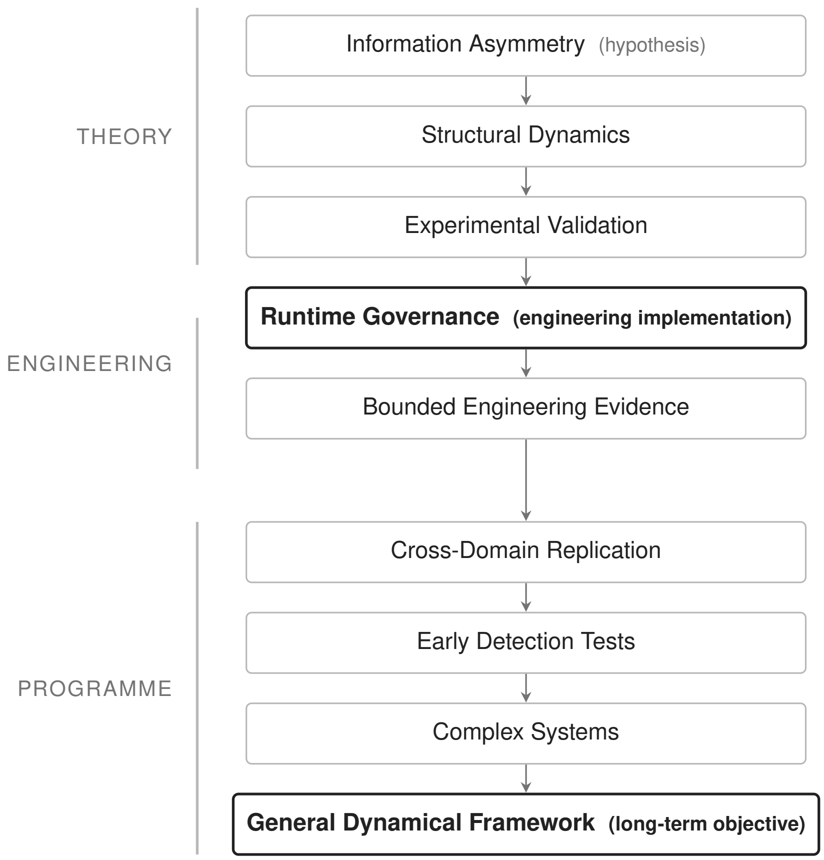


Figure 5. The research programme. The arrows show programme order, not proof. Runtime Governance is an engineering implementation motivated by structural concepts; scientific validation remains a separate empirical stage.

15 Interpretation and Implications

Taken together, the hypothesis suggests a layered view: psychological language may function as a compressed, readable surface, while dynamical structure specifies how the system evolves beneath that surface. The dashboard/engine metaphor is useful if treated carefully. A dashboard may preserve exactly the variables needed for one task and omit variables essential for another.

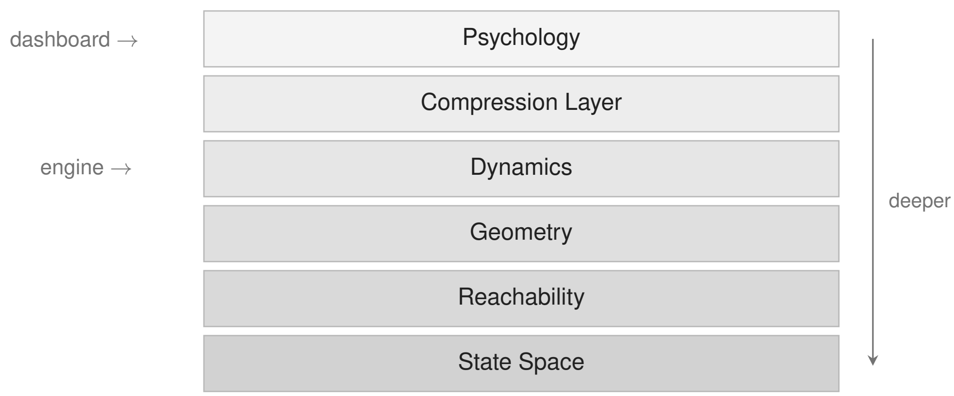


Figure 6. The compression hierarchy. Psychology is represented as a compact dashboard and dynamics as the generating engine. The diagram is a hypothesis about representational relation, not a settled biological hierarchy.

If the hypothesis is supported

The strongest implication would be methodological. Researchers could ask not only whether a label predicts an outcome, but which state-space distinctions the label merges and whether those distinctions improve early prediction or intervention. Psychological descriptions would remain useful macrovariables, while their limits would be mapped explicitly.

If the hypothesis is weakened

If labels prove sufficient for relevant future and intervention targets, this would show that carefully constructed psychological descriptions are more information-preserving than the hypothesis predicts. A null result could also reveal that dynamical measurements are too unstable, costly, or context-sensitive to improve on robust higher-level constructs.

What remains invariant

Either outcome improves the science because it replaces an argument about terminology with a comparison of representations. The programme is not committed to a favourable answer; it is committed to a test capable of returning one.

16 Future Research

The research should proceed in widening bands, with each transition conditioned on successful measurement and replication in the prior band.

PHASE I — MEASUREMENT

Synthetic ground truth · estimator validation · state reconstruction · preregistered distortions · negative controls

PHASE II — CONTROLLED BEHAVIOUR

Perturbation tasks · repeated measures · validated descriptions · intervention specificity · cross-participant transfer

PHASE III — EXTERNAL REPLICATION

New instruments · new tasks · distribution shift · competing state models · independent laboratories

PHASE IV — DOMAIN EXPANSION

Engineered systems first; biology and medicine only after validation; ecological and societal systems as long-term, separately governed programmes

Figure 7. A staged research roadmap. Expansion is conditional rather than automatic: each band requires evidence that the representation, metrics, and uncertainty estimates remain valid.

Priority research questions include:

– Which psychological constructs behave as near-sufficient macrovariables for specific targets, and which merge intervention-relevant dynamics?
– How stable is the asymmetry across timescales, individuals, cultures, instruments, and perturbation regimes?
– Can a low-dimensional dynamical representation preserve the predictive remainder without merely retaining all raw data?
– Which distortion measures best capture topology, reachability, resilience, and intervention rather than pointwise reconstruction?
– When do higher-level descriptions outperform lower-level detail by filtering noise and improving transportability?
– Can early-warning gains be independently replicated without leakage and without unacceptable false-alarm costs?
– What governance is required before any model is used in health, employment, education, or public decision making?

The long-term objective is not to force diverse disciplines into one vocabulary. It is to determine whether a shared set of dynamical primitives can support comparable questions about state, transition, constraint, reachability, resilience, and recoverability while respecting domain-specific mechanisms.

17 Long-Term Vision

The ambition is not merely to improve prediction within a single discipline. It is to investigate whether a common set of dynamical primitives can provide a comparable explanatory language across systems that are currently studied in isolation. Comparable does not mean identical: each domain retains its own state variables, mechanisms, ethical constraints, and validation standards.

Most disciplines act after a threshold is crossed — after the diagnosis, after the failure, after the collapse. A structural account offers a different posture. If the trajectories that precede catastrophic outcomes have a detectable geometry, they can be identified while the system is still recoverable. Rather than waiting for outcomes, the framework seeks the structure that precedes them.

If the hypothesis holds, related mathematics may eventually describe an autonomous agent before it acts, a biological system before decompensation, or an ecosystem before collapse — each as a system moving through a reachable space, approaching or avoiding a critical region. This is a long-term research possibility, not an inference licensed by the present engineering evidence. Each expansion must begin again with domain-specific measurement and falsification.

18 Conclusion

The central question is simple:

> *Can a dynamical description reproduce the explanatory and predictive role of psychological concepts across diverse situations, while also making accurate predictions that psychological labels alone cannot?*

This paper retains the central hypothesis unchanged: higher-level psychological descriptions are compressions of deeper dynamical descriptions. It strengthens the claim by narrowing its meaning, connecting it to adjacent literature without treating that literature as prior confirmation, and specifying the joint evidence required: forward expressivity, reverse underdetermination, predictive remainder, and intervention relevance.

The programme is deliberately asymmetric in its burdens. Predicting a label from rich data is not enough. A credible test must prevent leakage, match model capacity, quantify uncertainty, compare static and shuffled controls, verify perturbational consequences, and state observations that would weaken the proposal. It must also accept the possibility that some psychological descriptions are effective sufficient statistics for the targets that matter.

Runtime Governance remains an engineering implementation inspired by the structural vocabulary of state, trajectory, constraint, reachability, and forbidden regions. Its operational results demonstrate bounded engineering utility. They do not prove the scientific hypothesis. Maintaining that boundary is what allows both the engineering system and the research programme to be evaluated on their own evidence.

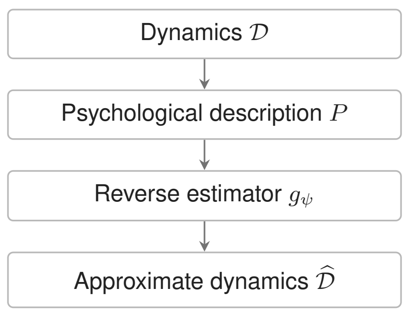
What begins as a question about psychology therefore becomes a disciplined comparison of representations: what each preserves, what each discards, which futures each predicts, and how early each can distinguish a recoverable trajectory from one approaching an irreversible outcome.

Appendix A Formal Summary

Let $\mathcal{D}$ be a dynamical representation, P a psychological description, and Z a preregistered target. The programme tests:

$$\begin{aligned}
&\textbf{Forward:} && \widehat{P} = f_\phi(\mathcal{D}), \\
&\textbf{Reverse:} && \widehat{\mathcal{D}} = g_\psi(P), \\
&\textbf{Reverse loss:} && L^*_{\text{rev}} = \inf_{g \in \mathcal{G}} \mathbb{E}[d_{\mathcal{D}}(\mathcal{D}, g(P))], \\
&\textbf{Predictive remainder:} && \Gamma_Z = I(\mathcal{D}; Z \mid P), \\
&\textbf{Sufficiency null:} && H_0 : \Gamma_Z \le \delta_Z \text{ and } L^*_{\text{rev}} \le \varepsilon_D.
\end{aligned}$$

The tolerances δ_Z and ε_D must be specified before confirmatory analysis and justified by practical rather than merely statistical significance.



$$\mathcal{D} \neq \widehat{\mathcal{D}} \quad \Longrightarrow \quad L^*_{\text{rev}} > \varepsilon_D \quad \text{for preregistered structure}$$

Reconstruction error alone is not decisive. It must be paired with forward adequacy and target-relevant remainder. A label that loses irrelevant microdetail may still be an excellent representation.

Appendix B Minimum Preregistration Record

Item	**Required declaration**
Domain and unit	System, participant, episode, timescale, inclusion criteria, and grouping structure.
Psychological representation	Instrument, rater procedure, reliability estimate, scoring, and permissible context.
Dynamical representation	Observations, actions, state model, training procedure, uncertainty, and label firewall.
Primary targets	Future event, perturbation response, transition, intervention, or topology property.
Distortion and thresholds	$d_{\mathcal{D}}$, equivalence margins, weights, and practical significance.
Baselines	Static, shuffled, capacity-matched, label-only, and raw-observation comparators.
Validation	Nested folds, system-held-out split, external task, and distribution-shift test.
Inference	Sample-size simulation, uncertainty interval, multiplicity control, and missing-data treatment.
Falsification	Exact pattern that weakens or fails to support the hypothesis.

References

Ashby, W. R. (1956). *An Introduction to Cybernetics.* Chapman & Hall.

Aubin, J.-P. (1991). *Viability Theory.* Birkhauser. doi:10.1007/978-0-8176-4910-4.

Beer, R. D. (2000). Dynamical approaches to cognitive science. *Trends in Cognitive Sciences*, 4(3), 91–99. doi:10.1016/S1364-6613(99)01440-0.

Berger, T. (1971). *Rate Distortion Theory: A Mathematical Basis for Data Compression.* Prentice-Hall.

Borsboom, D., Mellenbergh, G. J., and van Heerden, J. (2003). The theoretical status of latent variables. *Psychological Review*, 110(2), 203–219. doi:10.1037/0033-295X.110.2.203.

Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley. doi:10.1002/047174882X.

Cronbach, L. J., and Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, 52(4), 281–302. doi:10.1037/h0040957.

Crutchfield, J. P., and Young, K. (1989). Inferring statistical complexity. *Physical Review Letters*, 63(2), 105–108. doi:10.1103/PhysRevLett.63.105.

Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11, 127–138. doi:10.1038/nrn2787.

Gibson, J. J. (1979). *The Ecological Approach to Visual Perception.* Houghton Mifflin.

Haken, H., Kelso, J. A. S., and Bunz, H. (1985). A theoretical model of phase transitions in human hand movements. *Biological Cybernetics*, 51, 347–356. doi:10.1007/BF00336922.

Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1), 35–45. doi:10.1115/1.3662552.

Kelso, J. A. S. (1995). *Dynamic Patterns: The Self-Organization of Brain and Behavior.* MIT Press.

Littman, M. L., Sutton, R. S., and Singh, S. (2002). Predictive representations of state. In *Advances in Neural Information Processing Systems 14*, 1555–1561. proceedings.neurips.cc.

Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information.* W. H. Freeman.

Mitchell, I. M., Bayen, A. M., and Tomlin, C. J. (2005). A time-dependent Hamilton-Jacobi formulation of reachable sets for continuous dynamic games. *Journal of Scientific Computing*, 23, 203–237. doi:10.1007/s10915-004-4647-8.

Morrison, D. (2026a). *Geometric Control Theory of Cognition: A Reachability-Based Framework for Identity, Intelligence, and Experience.* github.com/davarntrades/morrison-framework.

Morrison, D. (2026b). *Morrison Runtime Governance: Pre-Execution Trajectory Governance for Autonomous Systems.* github.com/davarntrades/Morrison-Runtime-Governance.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

Resurrection Tech (2026a). *Runtime Governance — Live Demonstration.* resurrection-tech.com/live-demo. Accessed 10 July 2026.

Resurrection Tech (2026b). *Evidence and Methodology.* resurrection-tech.com/evidence. Accessed 10 July 2026.

Rissanen, J. (1978). Modeling by shortest data description. *Automatica*, 14(5), 465–471. doi:10.1016/0005-1098(78)90005-5.

Scheffer, M., Bascompte, J., Brock, W. A., Brovkin, V., Carpenter, S. R., Dakos, V., Held, H., van Nes, E. H., Rietkerk, M., and Sugihara, G. (2009). Early-warning signals for critical transitions. *Nature*, 461, 53–59. doi:10.1038/nature08227.

Shalizi, C. R., and Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104, 817–879. doi:10.1023/A:1010388907793.

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423; 27(4), 623–656. doi:10.1002/j.1538-7305.1948.tb01338.x.

Takens, F. (1981). Detecting strange attractors in turbulence. In D. Rand and L.-S. Young (eds.), *Dynamical Systems and Turbulence, Warwick 1980*, Lecture Notes in Mathematics 898, 366–381. Springer. doi:10.1007/BFb0091924.

Thelen, E., and Smith, L. B. (1994). *A Dynamic Systems Approach to the Development of Cognition and Action.* MIT Press.

Tishby, N., Pereira, F. C., and Bialek, W. (1999). The information bottleneck method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, 368–377. arXiv:physics/0004057.

van Gelder, T. (1998). The dynamical hypothesis in cognitive science. *Behavioral and Brain Sciences*, 21(5), 615–665. doi:10.1017/S0140525X98001733.

von Bertalanffy, L. (1968). *General System Theory: Foundations, Development, Applications.* George Braziller.

Wiener, N. (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine.* MIT Press.